University of Toronto Mississauga

Mathematical and Computational Sciences

MAT102H5F - Term Test 1

Test Booklet

Duration - 110 minutes

No Aids Permitted

Surname/Family Name: ______________________________

First/Given Name: ______________________________

Student Number: ______________________________

UTORid: ______________________________

Email: ______________________________

This exam contains 6 pages (including this cover page) and 5 problems. Check to see if any pages are missing and ensure that all required information at the top of this page has been filled in. You must also have the corresponding Multiple Choice Booklet, which contains the multiple choice questions needed for Question 5 (page 6). No aids are permitted on this examination. Examples of illegal aids include, but are not limited to textbooks, notes, calculators, or any electronic device.

Each part of Question 5 includes five multiple choice answers, of which only a single response is correct. All answers should be indicated on the included scantron sheet (page 6). When filling out the scantron:

- Use a #2 pencil to bubble answers (do not use ink). A #2 pencil ensures that the marks are dark enough.
- Ensure that bubbles are completely filled and make complete erasures.

Any failure to follow these instructions which results in the scanner not being able to read your work will result in a 3 mark deduction. You will not be penalized for incorrect answers.

Your solutions to Problems 1 - 4 must be written in the space provided. No additional paper may be submitted for grading. Do not, under any circumstances, write on the QR code in the top right corner of each page. **Doing so will result in you receiving a zero (0) on this test.**

Unless otherwise indicated, you are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work** in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work, will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- You may use any result mentioned explicitly in the readings, assignments, or in-class worksheets without proof. To use a result which is not covered by the aforementioned sources, you must provide a proof of the result as part of your work.

The Multiple Choice Booklet is printed one-sided, and may thus be used for rough work.

1. A set $A \subset \mathbb{R}$ is *bounded above* if there exists a $y \in \mathbb{R}$ such that for all $x \in A$, $x \leq y$.

   (i) (2 points) Give an example of a set which is bounded above.

   Solution: Set $A = \{1\}$. It's not too hard to see that by setting $y = 1$ (or any number larger than 1) we get that $x \leq 1$ for all $x \in A$.

   Rubric:

   - 1 point for a correct example.
   - 1 point for a correct argument that the example is in fact bounded above.

   (ii) (6 points) For a set $A \subset \mathbb{R}$ which is bounded above, we say that $y \in \mathbb{R}$ is an *upper bound* for A if for all $x \in A$, $x \leq y$. Are there any sets $A \subset \mathbb{R}$ with a unique upper bound? Either give an example of such a set A or show that there are no sets with a unique upper bound.

   Solution: There are no sets A with a unique upper bound. Indeed, suppose that A is a set which is bounded above, and that y is an upper bound for A. Then for all $x \in A$, $x \leq y \leq y+1$, and so $y + 1$ is also an upper bound for A. Since $y \neq y + 1$, the set A has at least two upper bounds.

   Rubric:

   - 1 point for correctly identifying that there are no such sets with unique upper bounds.
   - 4 points for a correct argument behind why upper bounds are not unique.
   - 1 point for a well written argument.

   (iii) (2 points) Using the definition of bounded above as a guide, define what it means for a set $A \subset \mathbb{R}$ to be *bounded below*.

   Solution: A set $A \subset \mathbb{R}$ is bounded below if there exists $y \in \mathbb{R}$ such that for all $x \in A$, $x \geq y$.

   Rubric:

   - 2 points for a correct statement.

2. (i) (3 points) Define the sets $A = (0, 6) \cap \mathbb{N}$, $B = \{x \in \mathbb{Z} : x \text{ is even}\}$, and $C = \{x \in \mathbb{Z} : x^2 \leq 4\}$. Compute $A \cap B, A \cap C$ and $B \cup C$, and very clearly show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution: We have $A = \{1, 2, 3, 4, 5\}$, $B = \{\dots, -4, -2, 0, 2, 4, \dots\}$ and $C = \{-2, -1, 0, 1, 2\}$. Thus

$$A \cap B = \{2, 4\}, \quad A \cap C = \{1, 2\}, \quad B \cup C = \{\dots, -6, -4, -2, -1, 0, 1, 2, 4, 6, \dots\}.$$

Now note that $(A \cap B) \cup (A \cap C) = \{1, 2, 4\}$. On the other hand, and similarly we have $A \cap (B \cup C) = \{1, 2, 4\}$. By inspection, these sets are equal as expected.

Rubric:

- 2 points for computing $A \cap B$, $A \cap C$ and $B \cup C$.
- 1 point for verifying the given identity.

(ii) (7 points) Show in general that if A, B, and C are general sets, then $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution: We proceed by double subset inclusion. Suppose first that $x \in A \cap (B \cup C)$, so that $x \in A$ and $x \in B \cup C$. The latter part in particular means that either $x \in B$ or $x \in C$. If $x \in B$ then $x \in A$ and $x \in B$ means that $x \in A \cap B$. If $x \in C$, then $x \in A$ and $x \in C$ means that $x \in A \cap C$.
In either case, we have that $x \in (A \cap B) \cup (A \cap C)$.

Conversely, suppose that $x \in (A \cap B) \cup (A \cap C)$. So that $x \in (A \cap B)$ or $x \in (A \cap C)$. In the former case, we have that $x \in A$ and $x \in B$, which means that $x \in A \cap (B \cup C)$. Similarly, if $x \in A$ and $x \in C$, then $x \in A \cap (B \cup C)$. Both cases give the same result, establishing the other inclusion.

Both inclusions give us the desired equality.

Rubric:

- 2 points for the $\subseteq$ direction.
- 2 points for the $\supseteq$ direction.
- 3 points for overall writing clarity.

3. Suppose that P, Q and R are predicates.

(i) (2 points) Write the truth table for $P \implies Q$.

Solution:
This is directly from the readings:

P	Q	$P \implies Q$
T	T	T
T	F	F
F	T	T
F	F	T

Rubric:

- 2 points for the table.

(ii) (8 points) Use truth tables to show that $(P \vee Q) \implies R$ is equivalent to $(P \implies R) \wedge (Q \implies R)$.

Solution:
We recall the truth tables for the $\vee$ and $\wedge$ operations:

P	Q	$P \vee Q$	$P \wedge Q$
T	T	T	T
T	F	T	F
F	T	T	F
F	F	F	F

Using these and part (a), we arrive at the truth tables for the two given statements:

P	Q	R	$P \vee Q$	$(P \vee Q) \implies R$
T	T	T	T	T
T	F	T	T	T
F	T	T	T	T
F	F	T	F	T
T	T	F	T	F
T	F	F	T	F
F	T	F	T	F
F	F	F	F	T

P	Q	R	$P \implies R$	$Q \implies R$	$(P \implies R) \wedge (Q \implies R)$
T	T	T	T	T	T
T	F	T	T	T	T
F	T	T	T	T	T
F	F	T	T	T	T
T	T	F	F	F	F
T	F	F	F	T	F
F	T	F	T	F	F
F	F	F	T	T	T

As the last column in both truth tables are identical, we conclude that the two given statements are logically equivalent.

Rubric:

- 6 points for a correct table, and 2 points for some level of description or explanation. Whether this is written explanation, uses auxiliary columns, or breaks the tables down into pieces.

4. For each of the below statements: (1) convert the statement to an English sentence, (2) say whether the statement is true or false, and (3) prove your answer.

   (i) (5 points) $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}, m + n$ is even.

   Solution: English: For all integers m, there exists an integer n such that $m + n$ is even. The statement is true.

   Fix $m \in \mathbb{Z}$. If m is even, then choose $n = 0$ so that $m + n = m$ is even. If m is odd, choose $n = 1$. Then by the definition of odd there exists an integer k so that $m + n = (2k + 1) + 1 = 2(k + 1)$, and so $m + n$ is even.

   Rubric:

   - 1 point for the English statement.
   - 3 points for the argument.
   - 1 point for overall writing quality.

   (ii) (5 points) $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m + n$ is even.

   Solution: English: There exists an integer m such that for all integers n, $m + n$ is even. This statement is false.

   We will prove the negation: $\forall m \in \mathbb{Z}, \exists n \in \mathbb{Z}, m + n$ is odd. Fix $m \in \mathbb{Z}$. If m is even, then choose $n = 1$ so that $m + n$ is odd. If m is odd, then choose $n = 0$ so that $m + n$ is odd.

   Alternatively, suppose for a contradiction that such a number m exists. This in turn means that both $m + 0 = m$ and $m + 1$ are even numbers. In the former case, we get that m must be an even number, which contradicts the latter case. **Rubric:**

   - 1 point for the English statement.
   - 3 points for the argument.
   - 1 point for overall writing quality.